

Linear transforms:

$$\bar{b} = A \cdot \bar{v}$$
$$\begin{matrix} m \times 1 & m \times n & n \times 1 \end{matrix}$$
$$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{m \times n} \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{n \times 1}$$

For each entry of $\bar{b}$, we have inner product of n length vectors.

Naive approach: $O(mn)$

$$L_{ij} = \omega^{ij}$$

(primitive n^{th} root of unity
 $\Rightarrow \omega^k \neq 1 \quad \forall k \in [1, n-1]$)

$$\omega = e^{\frac{2\pi i}{n}}$$

e.g. primitive fourth root
 $\omega = -i, i$
 $\omega^4 = 1, \omega^3 \neq 1, \omega^2 \neq 1, \omega \neq 1$

$$\begin{cases} \omega = i \\ \omega^2 = -1 \\ \omega^3 = -i \\ \omega^4 = 1 \end{cases}$$

If ω is the primitive fourth root, then
 ω^2 is the primitive 2nd root.

Vandermonde matrices

$$V(\alpha_1, \dots, \alpha_n) = \begin{bmatrix} \alpha_1^0 & \alpha_1^1 & \dots & \alpha_1^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_n^0 & \alpha_n^1 & \dots & \alpha_n^{n-1} \end{bmatrix}$$

$$P(z) = \sum_{i=0}^{n-1} c_i \cdot z^i$$

$$P(\alpha_i) = \langle \text{ith row of } V(\bar{\alpha}), \begin{bmatrix} c_0 \\ \vdots \\ c_{n-1} \end{bmatrix} \rangle \Rightarrow \begin{bmatrix} P(\alpha_1) \\ \vdots \\ P(\alpha_n) \end{bmatrix} = \underbrace{V}_{\text{given coeffs of a polynomial, we are computing their evaluations.}} \cdot \begin{bmatrix} c_0 \\ \vdots \\ c_{n-1} \end{bmatrix} \Rightarrow \sum_{j=0}^{n-1} c_j \cdot \alpha_i^j = P(\alpha_i)$$

coeffs:

$$\begin{bmatrix} c_0 \\ \vdots \\ c_{n-1} \end{bmatrix} = \underbrace{V^{-1}}_{\text{evaluations}} \cdot \begin{bmatrix} P(\alpha_1) \\ \vdots \\ P(\alpha_n) \end{bmatrix}$$

$$\begin{matrix} & V^{-1} \\ \text{Eval} & \longrightarrow & \text{Coefs} \\ & \longleftarrow V \end{matrix}$$

A polynomial of degree d can be obtained from its evals at $(d+1)$ points.

$$\begin{matrix} A(x) \cdot B(x) & = & C(x) \\ \downarrow & & \downarrow \\ \text{Degree: } d & & d \quad 2d \end{matrix}$$

$\therefore$ Evaluate $A(x)$ and $B(x)$ at $2d+1$ points
Multiply evals
 $\Rightarrow$ We get evals of $C(x)$ at $2d+1$ points
 $\Rightarrow$ We can find $C(x)$.

But still $O(d^2)$ $\therefore$ we use ω^i to obtain related values without actually multiplying.

ie. Complexity of linear transformation improves when $\alpha_1, \dots, \alpha_n$ are $\omega^1, \omega^2, \dots, \omega^n$.

We are interested in the complexity of linear transformations when $\alpha_1, \dots, \alpha_n$ are $\omega^1, \omega^2, \dots, \omega^n$.

$$(L^{-1})_{ij} = \frac{\omega^{-ij}}{n} = \frac{\omega^{n-ij}}{n}$$

If transformation by L and L^{-1} were efficient (much better than $O(n^2)$), then we can do polynomial mult. efficiently.

$$P(z) = \sum_{i=0}^n c_i \cdot z^{i-1} \quad Q(z) = \sum_{i=0}^n c'_i \cdot z^{i-1} \quad R(z) = P(z) \cdot Q(z)$$
$$\downarrow T(n) \quad \downarrow T(n) \quad \downarrow T'(n)$$
$$P(\omega), \dots, P(\omega^n) \quad Q(\omega), \dots, Q(\omega^n) \quad R(\omega), \dots, R(\omega^n)$$

(AWEAT: R need not be a deg n polynomial)

DFT(n) :

$$i \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix} \omega^{ij} \leftarrow \text{primitive } n^{\text{th}} \text{ root of unity.}$$

$$\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix} \omega^{ij} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

$$b_i = \sum_{j=0}^{n-1} \omega^{ij} \cdot a_j = \sum_{j=0}^{n/2} \omega^{ij} \cdot a_j + \sum_{j=\frac{n}{2}}^{n-1} \omega^{ij} \cdot a_j$$

$$\sum_{j=\frac{n}{2}}^{n-1} \omega^{ij} \cdot a_j = \sum_{j'=0}^{n/2-1} \omega^{i(j'+\frac{n}{2})} \cdot a_{j'+\frac{n}{2}} = \omega^{\frac{n}{2}i} \cdot \sum_{j'=0}^{n/2-1} \omega^{ij'} \cdot a_{j'+\frac{n}{2}}$$

$$\left\{ j' = j - \frac{n}{2} \Rightarrow j = j' + \frac{n}{2} \right\}$$
$$\omega^{\frac{n}{2}} = (-1)$$

$$\therefore b_i = \underbrace{\sum_{j=0}^{n/2-1} \omega^{ij} \cdot a_j}_{\text{If } \omega^n = 1, \text{ then } \omega^{n/2} = -1} + \omega^{\frac{n}{2}i} \cdot \sum_{j'=0}^{n/2-1} \omega^{ij'} \cdot a_{j'+\frac{n}{2}} = \sum_{j=0}^{n/2-1} \omega^{ij} \cdot a_j + (-1)^i \cdot \sum_{j'=0}^{n/2-1} \omega^{ij'} \cdot a_{j'+\frac{n}{2}}$$

Is this a $DFT_{\frac{n}{2}}$? NO, because ω is n^{th} root and not $\frac{n}{2}^{\text{th}}$ root.

OBSERVATION: ω^2 is $\frac{n^{\text{th}}}{2}$ primitive root of unity

① If i is even: Say $i = 2p$

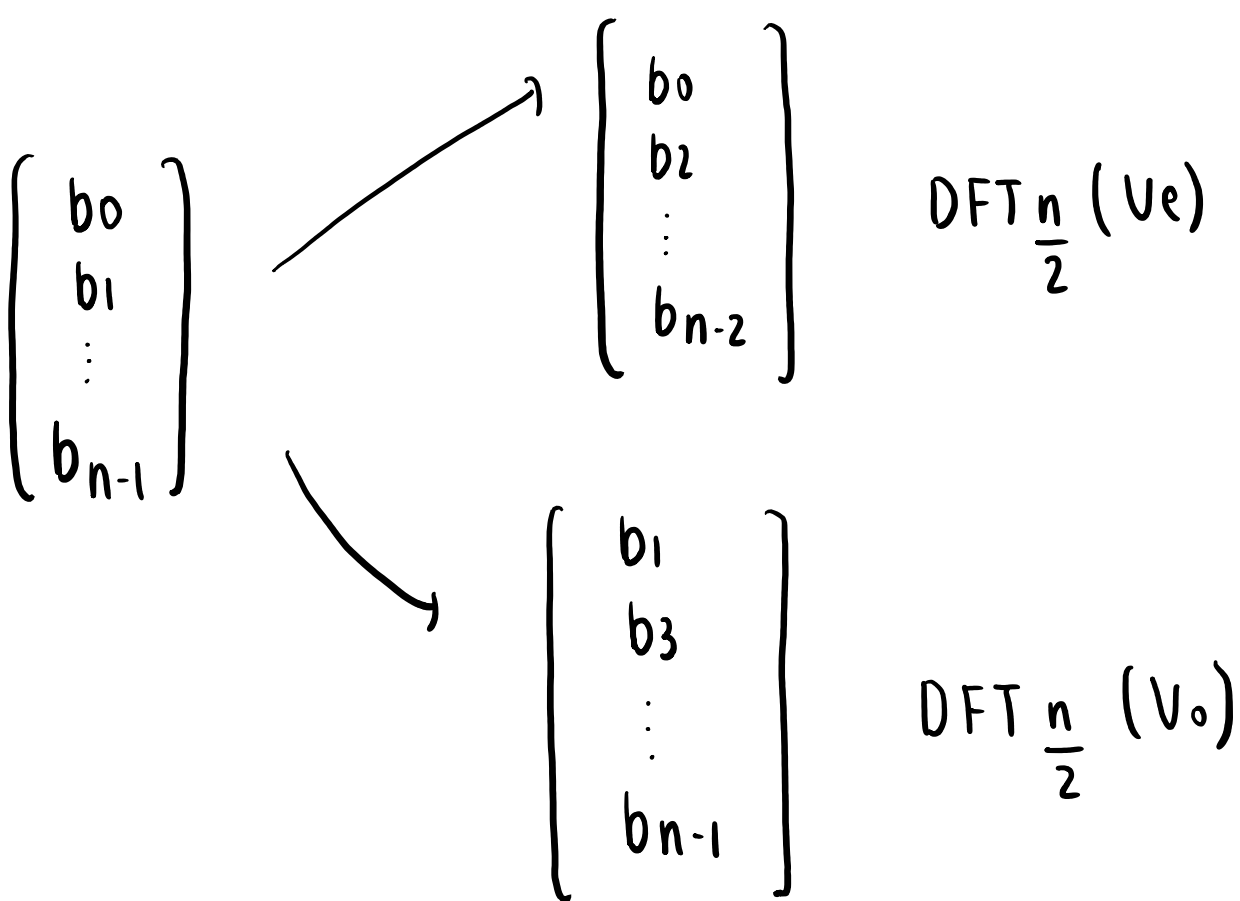
$$\Rightarrow b_i = \sum_{j=0}^{n/2-1} (\omega^2)^{pj} \cdot a_j + \sum_{j'=0}^{n/2-1} (\omega^2)^{pj'} \cdot a_{j'+\frac{n}{2}} \quad \left. \begin{matrix} j' \text{ is a notation} \\ \text{and can be replaced with } j. \end{matrix} \right\}$$
$$= \sum_{j=0}^{n/2-1} (\omega^2)^{pj} \cdot (a_j + a_{j+\frac{n}{2}})$$

② Else: $i = 2p+1$
 $ij = (2p+1)j = 2pj + j$

$$\Rightarrow b_i = \sum_{j=0}^{n/2-1} \omega^{2pj} \cdot a_j \cdot \omega^j - \sum_{j'=0}^{n/2-1} \omega^{2pj'} \cdot a_{j'+\frac{n}{2}} \cdot \omega^{j'}$$
$$= \sum_{j=0}^{n/2-1} (\omega^2)^{pj} \cdot \omega^j \cdot (a_j - a_{j+\frac{n}{2}})$$

$$V_e = \begin{bmatrix} a_0 + a_{\frac{n}{2}} \\ a_1 + a_{\frac{n}{2}+1} \\ \vdots \\ a_{\frac{n}{2}-1} + a_{n-1} \end{bmatrix} \quad V_o = \begin{bmatrix} a_0 - a_{\frac{n}{2}} \\ \omega (a_1 - a_{\frac{n}{2}+1}) \\ \vdots \\ \omega^{n/2-1} (a_{\frac{n}{2}-1} - a_{n-1}) \end{bmatrix}$$

$\therefore DFT_n(\bar{a})$ from $DFT_{\frac{n}{2}}(V_e)$ and $DFT_{\frac{n}{2}}(V_o)$



Fast (Discrete) Fourier Transform

- FFT(n)($\bar{a}$) :
- * Find primitive n^{th} root of unity.
 - * Construct vectors V_e and V_o
 - * Recursively call $FFT_{\frac{n}{2}}(V_e)$ and $FFT_{\frac{n}{2}}(V_o)$
 - * "Put together" the results
 - * Return $\bar{b}$ thus obtained.

$$T(n) = 2 \cdot T\left(\frac{n}{2}\right) + O(n) \Rightarrow O(n \log n)$$